Project Helios

Part 4: Risk Management & Budget

Mercury-based autonomous manufacturing for Dyson swarm construction

ENG 2001 - Project Management
York University
Fall 2025

Submitted: October 2025

Table of Contents

1. Risk Management Framework

Project Helios employs a structured risk management approach aligned with industry best practices and tailored to the unique challenges of autonomous space manufacturing on Mercury.

1.1 Risk Scoring Methodology

All risks are assessed using a 5×5 matrix evaluating Probability (P) and Impact (I) across four dimensions: Cost, Schedule, Technical Performance, and Safety & Compliance. The highest impact dimension determines the overall impact score.

Probability Scale (1-5):

- 1 - Very Low: ≤10% likelihood
- 2 - Low: 11-30% likelihood
- 3 - Medium: 31-50% likelihood
- 4 - High: 51-70% likelihood
- 5 - Very High: ≥71% likelihood

Impact Scale (1-5) by Dimension:

Cost Impact:

- 1: <1% over Estimate at Completion (EAC)
- 2: 1-5% over EAC
- 3: 5-10% over EAC
- 4: 10-20% over EAC
- 5: >20% over EAC

Schedule Impact (vs. Critical Path):

- 1: <1 week delay

- 2: 1-4 weeks delay
- 3: 1-2 months delay
- 4: 2-4 months delay
- 5: >4 months delay

Technical Performance Impact:

- 1: Negligible, within measurement noise
- 2: Minor rework required
- 3: Throughput/accuracy degraded 5-15%
- 4: Throughput/accuracy degraded 15-30%
- 5: >30% degradation or mission objective jeopardized

Safety & Compliance Impact:

- 1: Paperwork correction only
- 2: Minor non-conformance
- 3: Major QA finding, stand-down <2 weeks
- 4: Regulator intervention, stand-down >1 month
- 5: License revocation or uncontrolled behavior

Risk Score Calculation:
Risk Score = Probability × Impact

Exposure Bands:
• Low Risk: 1-6
• Medium Risk: 8-12
• High Risk: 15-25

1.2 Risk Response Strategies

Avoid: Eliminate the risk by changing project approach or scope

Mitigate: Reduce probability or impact through proactive measures

Transfer: Shift risk to third party (insurance, contracts, suppliers)

Accept: Acknowledge risk and prepare contingency response

1.3 Risk Categories

Project Helios risks are organized into three primary categories aligned with the technical architecture and operational environment:

Technical/Replication & Manufacturing: Risks related to ISRU throughput, additive/subtractive manufacturing quality, replication quality assurance, and autonomous assembly processes.

Environmental: Mercury surface hazards including extreme thermal cycles (-170°C to +430°C), cumulative radiation exposure, abrasive dust contamination, and micrometeoroid impacts.

Systemic/Operational & Governance: Mission operations risks covering EDL, mass-driver launches, communications, supply chain constraints, regulatory compliance, and public acceptance.

2. Detailed Risk Register

The following comprehensive risk register identifies 20 key risks across all project phases, with detailed analysis of causes, triggers, mitigation strategies, and contingency plans. Risk ownership is assigned to roles established in Part 3.

2.1 Technical/Replication & Manufacturing Risks

ID	Risk	Cause & Triggers	P	I	Score	Owner	Response	Contingency
R-01	Replication errors propagate	AM dimensional drift; assembly misalignment; QA yield <99%. Trigger: >0.5% non-conformance per 20 units	3	5	**15-H**	Test & QA + Autonomy	MITIGATE: First-article inspection; HIL tests; import avionics early; behavioral diversity	Quarantine batch; rollback firmware; slow cadence until yield recovers
R-02	Uncontrolled replication	Kill-switch failure; command spoofing. Trigger: replication count exceeds cap window	2	5	10-M	Safety & Ethics	AVOID/MITIGATE: Hard-coded caps; authenticated kill-switch; resource quotas; auditable logs	External abort; de-energize replication cells
R-03	ISRU throughput shortfall	Beneficiation/reduction underperform. Trigger: <0.5 t/month by M-04	3	4	12-M	ISRU Lead	MITIGATE: Parallelize lines; tune thermal budgets; sensor-based control; buffer feedstock	Interim Earth-supplied kits; defer non-critical fabrication
R-04	AM/SM defects & material brittleness	Regolith-derived alloys variable; inadequate sinter. Trigger: scrap rate >5%	3	4	12-M	Manufacturing Lead	MITIGATE: Process qualification; in-process sensing; post-process heat-treat; conservative allowables	Rework inventory; reprint with adjusted parameters
R-05	Software/ML failure or drift	Unseen states; coordination conflicts. Trigger: rising task collisions; watchdog resets	2	4	8-M	Autonomy Lead	MITIGATE: Formal verification for safety logic; sandboxed OTA updates; multi-agent conflict resolution	Partition network; roll back release

2.2 Environmental Risks

ID	Risk	Cause & Triggers	P	I	Score	Owner	Response	Contingency
R-06	Thermal extremes	Long solar day/night; poor thermal margins. Trigger: component temps near limits	3	4	12-M	Chief Systems Eng	MITIGATE: Rad-hard/thermally-shielded electronics; duty-cycle tasks; thermal storage; sun-safe parking	Swap units; reschedule ops to thermal windows

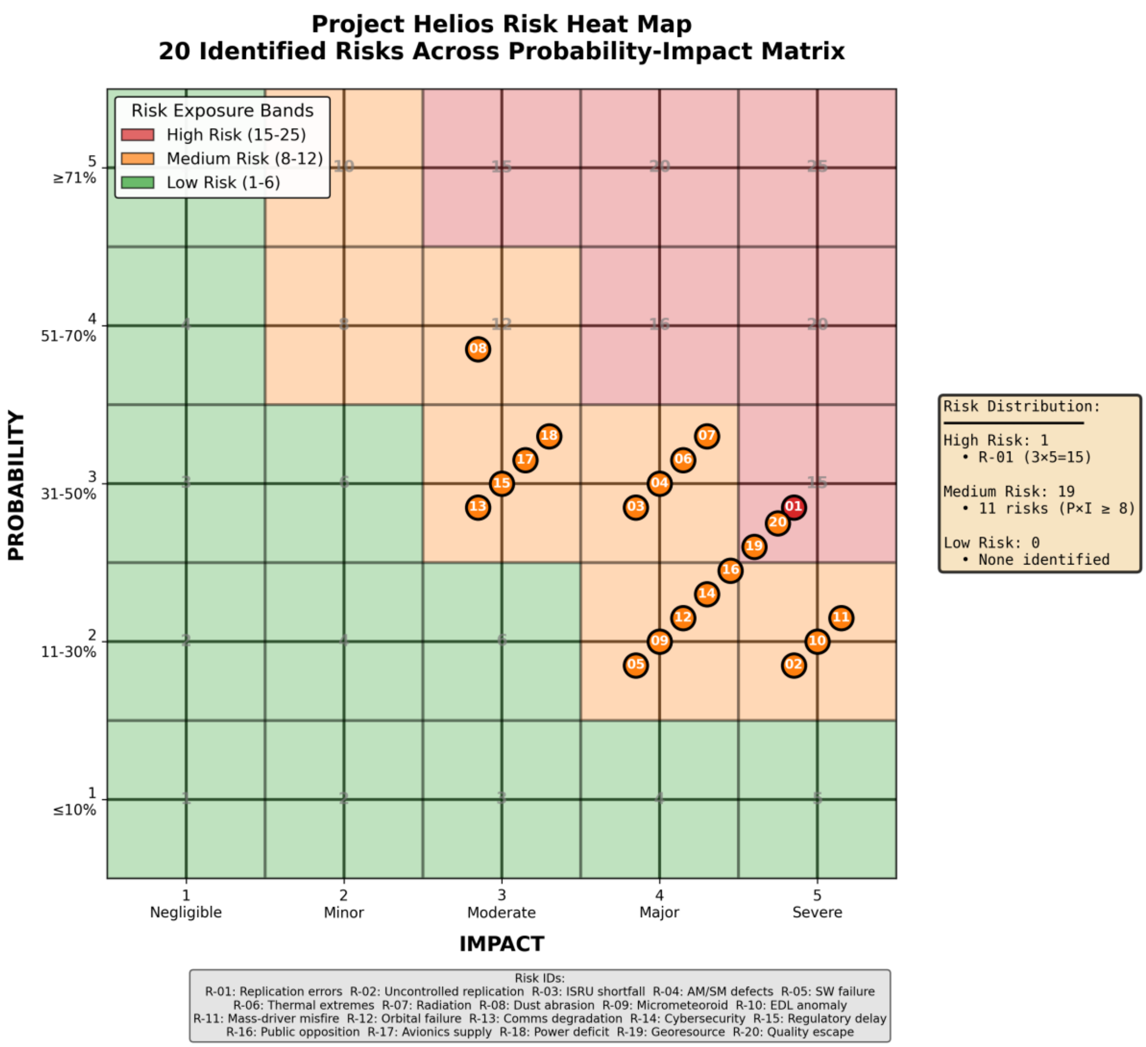
R-07	Radiation degradation	Cumulative TID/SEE. Trigger: dose meters trending above spec	3	4	12-M	Chief Systems Eng	MITIGATE: Rad-hard parts; shielding; ECC and redundancy; graceful degradation	De-rate performance; replace boards
R-08	Dust abrasion/contamination	Abrasive fines enter joints/AM chambers. Trigger: torque increase, filter pressure rise	4	3	12-M	SysEng + Mfg	MITIGATE: Seals/purges; dust-repellent coatings; filtered process gas; hardened bearings	Spare kits; accelerated lube/clean cycles
R-09	Micrometeoroid/impact damage	Unshielded surfaces. Trigger: impact sensor events; leak/breach alarms	2	4	8-M	SysEng	MITIGATE: Whipple shields on exposed assets; distributed redundancy	Isolate/power-down damaged modules

2.3 Systemic/Operational & Governance Risks

ID	Risk	Cause & Triggers	P	I	Score	Owner	Response	Contingency
R-10	EDL/landing anomaly	GN&C off-nominal; terrain hazards. Trigger: dispersion outside ops envelope	2	5	10-M	Launch Ops Lead	AVOID: End-to-end sims; terrain-relative nav; multi-site landing maps	Operate from alternate site; deploy relay masts
R-11	Mass-driver misfire/trajectory errors	Coil timing/power fault; structural resonance. Trigger: off-spec velocity/exit angle	2	5	10-M	Launch Ops + Mfg	MITIGATE: Staged power ramps; interlocks; high-fidelity FEA; prove-out to M-06	Stand-down; revert to solar-sail assist
R-12	Orbital injection/aggregation failure	Navigation/attitude errors. Trigger: tracking loss; ΔV margins eroded	2	4	8-M	Orbital Ops Lead	MITIGATE: GPS-independent nav aids; robust beacons; conservative ΔV; ground tracking	Retrieve/re-task modules; replace lost inventory
R-13	Mesh comms degradation	Congestion; line-of-sight loss. Trigger: packet loss increase, latency increase	3	3	9-M	Autonomy Lead	MITIGATE: Store-and-forward; TDMA windows; redundant bands; operate-alone fallbacks	Static beacon relays; pre-planned time-slot ops
R-14	Cybersecurity/firmware compromise	Unsigned updates; supply-chain poisoning. Trigger: integrity check fails	2	4	8-M	Autonomy + QA	MITIGATE: Signed/attested updates; SBOM; isolated staging; diversity in algorithms	Partition network; golden image restore
R-15	Regulatory & governance delay	UNOOSA/COSPAR concerns; debris policy. Trigger: extended review cycles	3	3	9-M	PM + Sponsor	MITIGATE: Early engagement; transparent logs; environmental & debris plans	Rephase on-surface build; reduce launch tempo
R-16	Public opposition/ethics backlash	Replication fears; space environment impact. Trigger:	2	4	8-M	Comms + Safety & Ethics	MITIGATE: Transparent governance; publish	Crisis comms; pause replication

		negative media trend					replication logs; independent audits	cadence
R-17	Imported avionics supply constraint	Lead-time spikes; export issues. Trigger: PO slips; supplier risk alerts	3	3	9-M	Supply Chain Lead	TRANSFER/MITIGATE: Dual-source; buffer stock; design alternates; long-lead funding	Re-prioritize replication scope; slower cadence
R-18	Power deficit during bootstrap	Array/concentrator lag; storage shortfall. Trigger: power margin <10%	3	3	9-M	SysEng + ISRU	MITIGATE: Prioritize power field build-out; energy-aware scheduling; thermal storage	Shed non-critical loads; import RTG backup
R-19	Georesource mischaracterization	Ore grade/terrain worse than expected. Trigger: survey variance >20%	2	4	8-M	ISRU Lead	MITIGATE: Multi-sensor survey; pilot pits; adaptive site selection	Shift pit locations; accept lower throughput temporarily
R-20	Quality system escape	QA/QC process gap. Trigger: post-delivery defect found in field	2	4	8-M	Test & QA Lead	MITIGATE: CDR/ORR gates; FMEA on replication process; configuration control	Field retrofit; CAR/PAR cycle

2.4 Risk Heat Map Summary



The risk portfolio shows one High-risk item (R-01: Replication errors) requiring immediate executive attention, with 19 Medium-risk items under active management. This distribution reflects the technical uncertainty inherent in early-phase autonomous space manufacturing and the extreme Mercury environment.

Risk Distribution:

• High Risk (15-25): 1 risk
• Medium Risk (8-12): 19 risks
• Low Risk (1-6): 0 risks

Note: The absence of low-risk items reflects conservative assessment given TRL gaps and Mercury environmental challenges.

3. Cost Estimate and Basis of Estimate

3.1 Total Project Cost

Total Baseline: $170 Billion USD
Project Duration: 20 years (Y1-Y20)
Currency: United States Dollar (USD)

3.2 Cost Distribution Principle

The Project Helios cost structure follows a front-loaded investment model where initial seed payload development and Mercury delivery dominate total expenditure. Once autonomous replication and ISRU systems are operational, incremental costs drop dramatically as the system becomes self-sustaining, requiring only Earth-based mission control and governance oversight.

Front-Loaded Investment: Seed payload design, integration, and Mercury delivery account for approximately 47% of total project cost, concentrated in Years 1-3.

Autonomous Phases: Bootstrap replication, infrastructure expansion, and module production incur minimal incremental cost beyond initial seed investment and ongoing Earth-based oversight.

Operational Tail: Mission control, governance, compliance, and spare parts costs spread across Years 6-20 at approximately $1B per year.

3.3 WBS-Aligned Cost Breakdown

WBS	Work Package	Cost (USD B)	Contingency %	Notes
1	Program Mgmt & Governance	8	15%	Compliance, QA, ethics oversight
2	Seed Payload (design, integration)	42	20%	High TRL gap; autonomy stack; AM/ISRU prototypes
3	Mercury Delivery & Commissioning	38	15%	Multiple launches; trajectory ops; landing systems
4	Bootstrap ISRU & Manufacturing	26	20%	Initial ISRU lines; AM/SM setup; redundancy
5	Controlled Self-	12	20%	QA/HIL;

	Replication			governance enforcement; imported avionics
6	Infrastructure Expansion	24	20%	Mass-driver prototype; power fields; logistics
7	Module Production & Deployment	40	15%	Fabrication ramp-up; orbital injection; tracking
8	Operations & Maintenance	20	10%	Mission control; compliance reporting; spares
	TOTAL	**170**	**—**	**Baseline from Part 1**

3.4 Basis of Estimate Narrative

The $170B baseline originates from the Part 1 business case and is allocated across WBS elements using a top-down approach, refined by the following drivers:

Front-Loaded Cost Drivers: Seed Payload Development ($42B) and Mercury Delivery & Commissioning ($38B) together account for 47% of total cost. These packages cover autonomy stack development, ISRU/AM prototypes, environmental hardening, QA systems, launch vehicles, trajectory operations, and EDL systems.

Technology Readiness Level Risk: Higher contingency percentages (20%) applied to lower-TRL elements: ISRU and additive manufacturing in space (TRL 3-5), mass-driver prototype (TRL 4-6), and replication governance systems. Mature elements like mission control receive lower contingency (10-15%).

Autonomous Scaling Economics: Once seed payload is operational, replication and module fabrication leverage in-situ resources and controlled self-replication, incurring negligible incremental cost beyond Earth-based QA oversight and governance monitoring.

Operational Steady State: $20B spread over 14 years (Y7-Y20) for mission control, compliance, and spares assumes no catastrophic failures and successful transition to autonomous operations.

3.5 Key Cost Assumptions

- Currency baseline in USD; optional conversion to CAD for reporting
- Heavy-lift rockets for seed payload delivery; electromagnetic mass-driver for all Mercury-to-orbit deployments
- Early generations import microelectronics and complex avionics; transition to higher local content by Year 6
- Autonomous scalability: replication and ISRU expansion occur without additional launches except contingency shipments
- Earth-based mission control at ~$1B per year post-Year 6, including governance, QA audits, and compliance reporting
- Contingency logic applied at WBS level based on TRL and environmental risk; excludes catastrophic failure scenarios (addressed in risk register)
- Entire Mercury operations are robotic; no crewed missions factored into cost structure

3.6 Cost Risk Drivers

Several high-impact risks could trigger contingency drawdown or require additional Earth shipments, potentially increasing total cost by 5-10%:

- R-01 (Replication errors propagating) may require extended QA cycles and slower replication cadence
- R-03 (ISRU throughput shortfall) could necessitate interim Earth-supplied component kits
- R-11 (Mass-driver misfire/trajectory errors) may force fallback to less efficient launch methods
- R-15 (Regulatory and governance delays) could extend program duration by 1-2 years, adding $1-2B in overhead per year of slip

4. Detailed Time-Phased Budget

The time-phased budget demonstrates the front-loaded investment strategy, with peak expenditures during Years 1-5 as seed payload development, Mercury delivery, and bootstrap ISRU/replication systems are established. Costs taper significantly in Years 6-20 as autonomous operations take over.

Year	Key Activities & Milestones	Annual Cost (USD B)	Cumulative (USD B)	Primary Cost Drivers
Y1	Seed design & integration; autonomy stack; governance plan (M-01: CDR prep)	**24**	24	Autonomy integration, ISRU/AM prototypes, QA systems
Y2	Launch & EDL; Mercury landing; site survey (M-02: Launch, M-03: Landing)	**33**	57	Heavy-lift launch vehicles, trajectory ops, EDL systems
Y3	ISRU commissioning; AM/SM setup; replication caps (M-04: ISRU operational)	**36**	93	Processing lines, manufacturing cells, replication governance
Y4	G1 replication; scale-out mining (M-05: First controlled replication)	31	124	First-generation fleet commissioning, mining expansion
Y5	Infrastructure expansion; power fields; logistics	31	155	Mass-driver development, power arrays, site infrastructure
Y6	Mass-driver prototype fired; QA validation (M-06 achieved)	1	156	Earth-based oversight only; autonomous ops established
Y7	Module fabrication; first orbital launches (M-07: First	1	157	Earth-based oversight; autonomous fabrication and

	orbit)			launch
Y8	Production cadence growth; orbital aggregation	1	158	Ramp to 100 modules/month baseline
Y9-Y20	Operations & Maintenance (mission control, compliance, governance)	12	170	Steady-state ops at ~$1B/year; no additional launches

4.1 Budget Allocation Rationale

Years 1-3 (Bootstrap Phase): Heavy investment in seed payload development, autonomy stack, ISRU prototypes, and QA systems. Launch and EDL costs peak in Y2. ISRU and AM/SM commissioning dominate Y3. These years consume ~55% of total budget establishing all enabling infrastructure.

Years 4-5 (Scale-Out & Infrastructure): Controlled replication and mining expansion require QA and governance oversight but minimal new hardware imports. Mass-driver prototype development and power field build-out drive Y5 costs.

Years 6-8 (Deployment Ramp-Up): Mass-driver testing complete; module fabrication and orbital launches begin. Costs drop sharply as replication and ISRU handle production autonomously with Earth-based oversight only.

Years 9-20 (Operational Tail): $1B per year for mission control, compliance, and spares, assuming no catastrophic failures. Total operational tail cost of $12B.

5. Status Budget Report (End of Year 3)

Report Date: End of Year 3

Overall Status: YELLOW (AMBER)

Cost and schedule variances require corrective action, but overall scope and quality remain intact.

5.1 Earned Value Analysis

Metric	Value (USD B)
Planned Value (PV)	93
Earned Value (EV)	88
Actual Cost (AC)	96
Cost Performance Index (CPI)	**0.92**
Schedule Performance Index (SPI)	**0.95**
Schedule Variance (SV)	−5
Cost Variance (CV)	−8

Interpretation:

- CPI = 0.92 (below 1.0) indicates cost overrun; project is over budget
- SPI = 0.95 (below 1.0) indicates slight schedule slip; project is behind schedule
- Earned Value of $88B represents ~94.6% of planned work completion

5.2 Variance Analysis

Schedule Variance (SV = −$5B)

Root Cause: ISRU throughput lag during commissioning; payload-centric assembly routines required additional hardware-in-the-loop iterations.

Impact: Minor slip on G1 commissioning start date. Year 4 Quarter 1 target still achievable with schedule compression techniques.

Cost Variance (CV = −$8B)

Root Cause: Unplanned expenses for enhanced thermal shielding, dust-proofing measures, and higher-than-expected first-article scrap rates during AM/SM commissioning.

Impact: CPI below 1.0 creates risk of exceeding $170B baseline if uncorrected. Immediate corrective action required.

5.3 Corrective Actions

ISRU Bottleneck Resolution: Deploy additional parallel beneficiation skid; optimize thermal budget allocation; implement advanced sensor-based process control.

Dust/Thermal Mitigation: Update supplier specifications for improved environmental protection; accelerate deployment of dust-repellent coatings and enhanced sealing systems.

QA Overhead Management: Continue use of imported avionics for one additional G1 production lot to reduce rework risk and improve first-pass yield.

Governance Verification: Confirm zero uncontrolled replication events; validate all replication caps and kill-switch mechanisms functioning as designed.

5.4 Forecast and Recovery Plan

Estimate at Completion (EAC): $172B (CPI-based projection)

Recovery Strategy:
- Trim non-critical spare parts inventory
- Defer optional test series to Year 4 without impacting critical path
- Negotiate favorable rates on long-lead imported components
- Accelerate autonomous operations transition to reduce Earth-based oversight costs

Target: Maintain baseline of $170B through efficiency improvements and scope optimization.

5.5 Status Dashboard Summary

Category	Status
Overall Project Status	AMBER
Scope	GREEN
Schedule (SPI 0.95)	AMBER
Budget (CPI 0.92)	AMBER
Quality	GREEN
Top Active Risks	R-01, R-03, R-04, R-08

Upcoming Critical Milestones:
- M-05: G1 commissioning (Year 4, Quarter 1)
- M-05: Scale-out mining operational (Year 4, Quarter 4)
- M-06: Mass-driver prototype fired (Year 6, Quarter 3)
- M-07: First orbit modules deployed (Year 7, Quarter 4)

6. Appendices

6.1 Appendix A: Risk Scoring Legend

Complete probability and impact scales with exposure band calculations are provided in Section 1.1 Risk Scoring Methodology.

6.2 Appendix B: Mapping to Previous Deliverables

Part 1 (Charter & Business Case):
Governance and ethics framework → R-02 (Uncontrolled replication), R-15 (Regulatory delay), R-16 (Public opposition)
$170B baseline establishes Cost Estimate foundation

Part 2 (WBS & Milestones):
ISRU commissioning activities → R-03
Mass-driver prototype development → R-11
Orbital deployment operations → R-12
Milestones M-02 through M-07 drive time-phased budget allocations

Part 3 (QA & Roles):
Replication caps and kill-switch protocols → R-02
QA gates and HIL testing requirements → R-01, R-20
Roles matrix establishes risk ownership assignments

Technical Report:
TRL gaps inform cost contingency logic
Mercury thermal extremes and radiation environment → R-06, R-07
Dust abrasion considerations → R-08
Payload-centric autonomy and resilience architecture → R-05, R-13
Ethical safeguards and governance → R-02, R-16

6.3 Appendix C: Key Project Assumptions

8. 1. Launch architecture uses electromagnetic accelerator (mass-driver) exclusively for Mercury-to-orbit deployments; Mercury has no atmosphere
9. 2. Cost structure is front-loaded with seed payload development and Mercury delivery accounting for ~47% of total expenditure
10. 3. Post-bootstrap phases incur negligible incremental cost beyond Earth-based oversight due to autonomous scalability
11. 4. Early generations import microelectronics and complex avionics; transition to higher local content by Year 6

12. 5. Operational steady-state costs $1B per year for mission control and compliance from Year 6 onward
13. 6. Governance framework includes mandatory phase-gates (CDR, ORR) before replication cadence increases or mass-driver payload launches commence
14. 7. All cost estimates in USD baseline; optional conversion to CAD for reporting purposes
15. 8. No human presence on Mercury; entire operation is robotic and autonomous

6.4 Appendix D: Compliance & Governance Framework

International Treaties: Outer Space Treaty, COSPAR planetary protection protocols

Ethics Standards: Transparent replication logs, authenticated kill-switches, resource quotas per generation

Quality Assurance: Phase-gate reviews (CDR, ORR), configuration control, FMEA, formal verification

Quality Control: First-article inspections, non-destructive evaluation, hardware-in-the-loop testing

Audit Requirements: External review for replication governance compliance and environmental impact assessment

7. References

[1] Y. Kaushal, "Self-Replicating Autonomous Systems for Scalable Infrastructure in Space Megaprojects," unpublished technical report, ENG 2003 – Effective Engineering Communication, Jul. 29, 2025. [Online]. Available: https://yathharthha.space/assets/docs/Kaushal_Yathharthha_TermProject1_Phase4_ENG2003.pdf